

Evan Liu

Boston, MA | eliu23@bu.edu | 206-334-3224 | <https://github.com/evanl23>

Education

Boston University | Boston, MA

May 2027

B.A. in Economics and Computer Science

- 3.73 GPA (Dean's List)
- Division 1 athlete on BU Men's Swim and Dive Varsity program
- **Relevant Coursework:** Operating Systems, Algorithms, Database Management Systems

Relevant Experience

Azabu Insights | Tokyo, Japan

June 2025 – Sep 2025

Software Engineer

- Engineered production-ready AI voice agents using OpenAI Whisper + GPT APIs to automate patient intake, reducing average onboarding time for clinics in Japan by 50%
- Designed and implemented responsive UI components in React + Tailwind with accessible layouts and transitions, improving usability across desktop and mobile

Finance & Investment Club | Boston University

Sep 2023 – Dec 2024

Senior Quantitative Researcher / Developer

- Built and backtested multiple algorithmic trading strategies with MACD and Bollinger Bands indicators
- Refined trading algorithms by training on past financial data and ultimately implementing on recent pricing to calculate potential returns
- Analyzed risks of stock portfolios by examining size, value, and market risk factors by applying the Fama French Factor Model

Projects

SMS-Based AI Productivity Assistant

github.com/evanl23/AI-Texting-Assistant

(200+ users | \$5,000 in BU funding)

- Engineered a Dockerized Flask backend hosted on Google Cloud Run, integrating Twilio webhooks for SMS interactions, OpenAI APIs for background NLP processing, and Firestore for persistent user data storage
- Implemented advanced natural language understanding with intent classification, fuzzy string matching, and personalized memory features, enabling agentic, context-aware conversations for 200+ users
- Integrated Google OAuth 2.0 using google-auth libraries to securely authenticate users and manage token lifecycles for Gmail and Calendar integrations
- Selected for BU's Innovative Pathway Program and secured \$5,000 in funding

Cloud-Hosted Raman Spectroscopy Query System

github.com/evanl23/RamanSpectroscopy

- Deployed a serverless FastAPI backend on AWS ECS Fargate to process and query Raman spectroscopy data captured by a detachable iPhone device for chemical substance identification
- Containerized the backend using Docker and pushed production-ready images to AWS ECR; used Qdrant vector database for similarity search on Raman spectra embeddings (3K+ spectra indexed)
- Configured scalable AWS infrastructure with ALB and VPC to enable public access and route HTTP requests for real-time querying of chemical signatures

Memory Probing Program in Real and Protected Mode

- Built a bare-metal x86 operating system in assembly and C that probes system memory using BIOS interrupts in Real Mode and GRUB multiboot memory mapping in Protected Mode
- Implemented memory management routines that parse hardware memory maps, calculate available RAM, and display memory regions with type classification and address ranges in both 16-bit and 32-bit modes
- Wrote low-level system software using AT&T assembly and C, including custom video memory drivers, ACPI E820 memory detection routines, and bootloader integration without external libraries